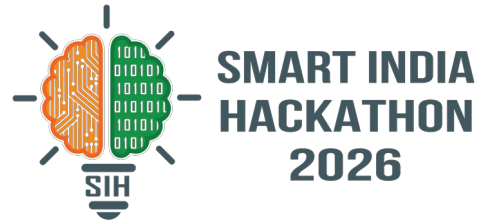


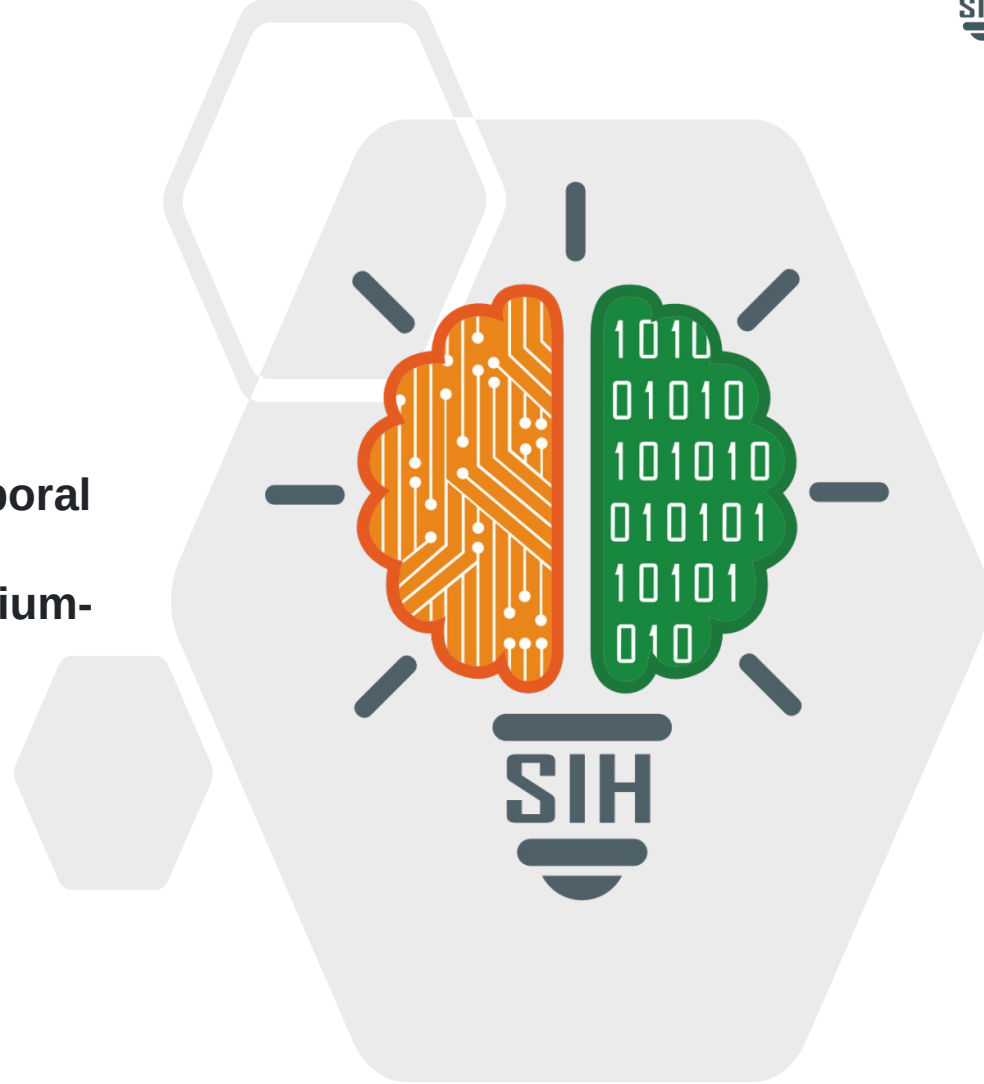
SMART INDIA HACKATHON 2026



MEGHA-DRISHTI मेघ-दृष्टि

AI-driven tracking and peak-preserving downscaling of extreme weather

- Problem Statement ID – 26078
- Problem Statement Title- AI-Driven Spatio-Temporal Tracking of Extreme Weather Anomalies in Medium-Range Forecasts
- Theme- Smart Automation
- PS Category- Software
- Team ID- 159951
- Team Name (Registered on portal) - CodeX_2026



MEGHA-DRISHTI : AI-driven tracking and peak-preserving downscaling of extreme weather in medium-range forecasts

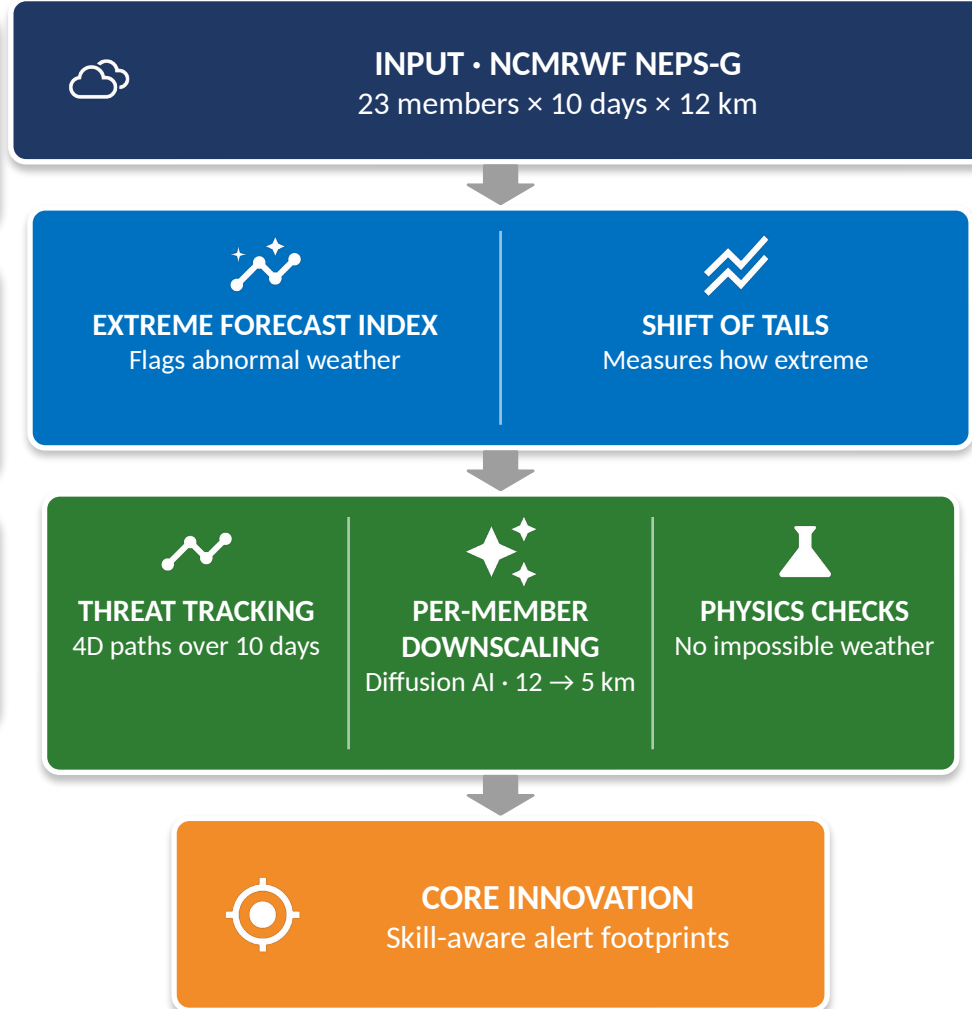
Real-world issue: India's 4 km ensemble forecast runs only 3 days ahead. Days 4–10 have only 12 km, where local extremes get lost.

Why important: Standard AI weather models blur storm and heat peaks — and averaging an ensemble blurs them too.

Solution: Auto-detect extreme threats in NCMRWF's 23-member ensemble, track them, and sharpen every member to ~5 km.

[Website · Video](#)

Prototype



Problem ↔ **Our Fix**

Peaks averaged away

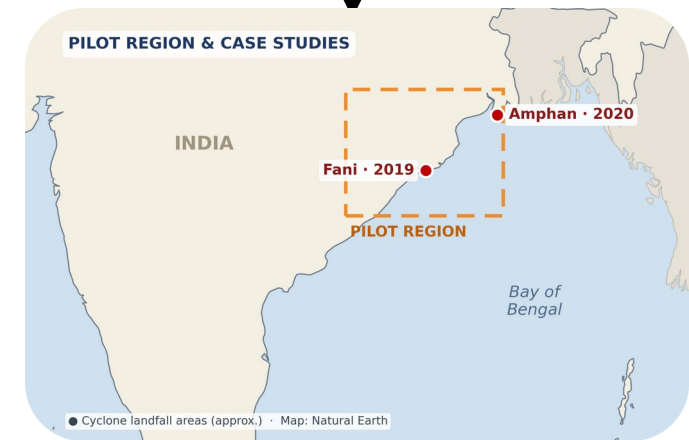
Diffusion AI per member

Manual chart scanning

Automated threat tracks

Over-precise false alerts

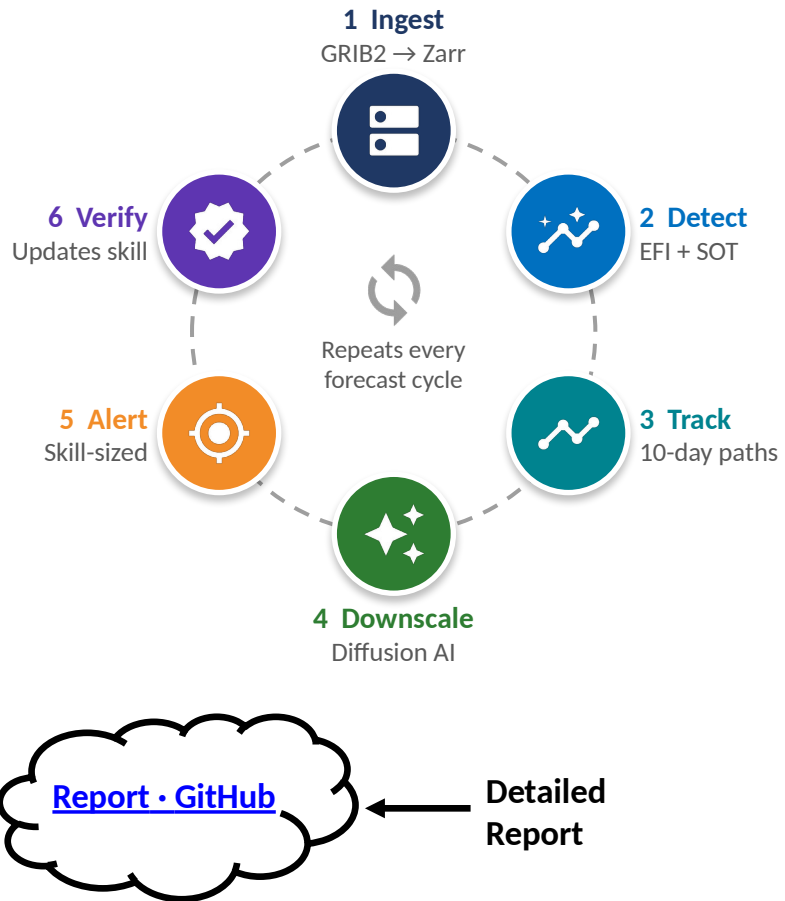
Skill-sized footprints



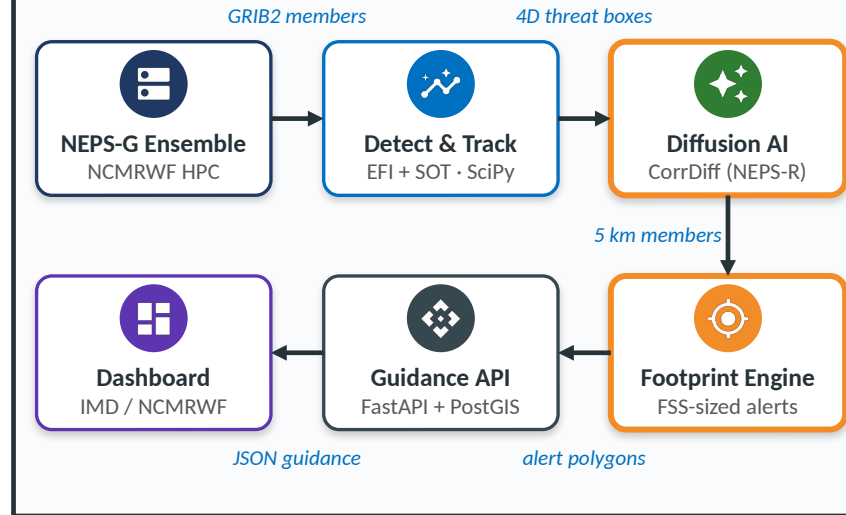
Sources: NCMRWF/WMO training, 2022 · Price et al., Nature 2025 · ECMWF Forecast User Guide §8.1.9

TECHNICAL APPROACH

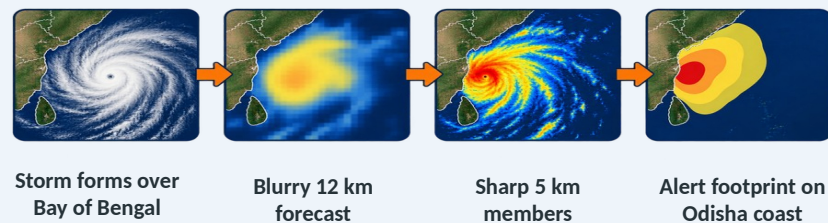
METHODOLOGY & PROCESS OF IMPLEMENTATION



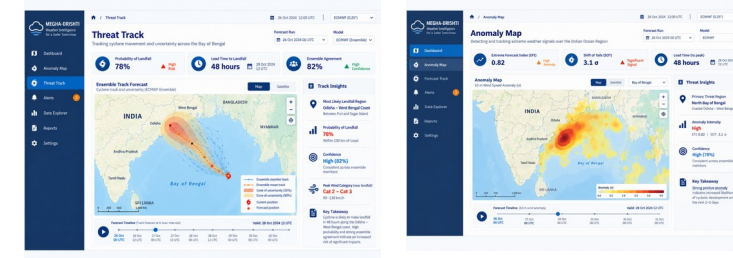
SYSTEM ARCHITECTURE



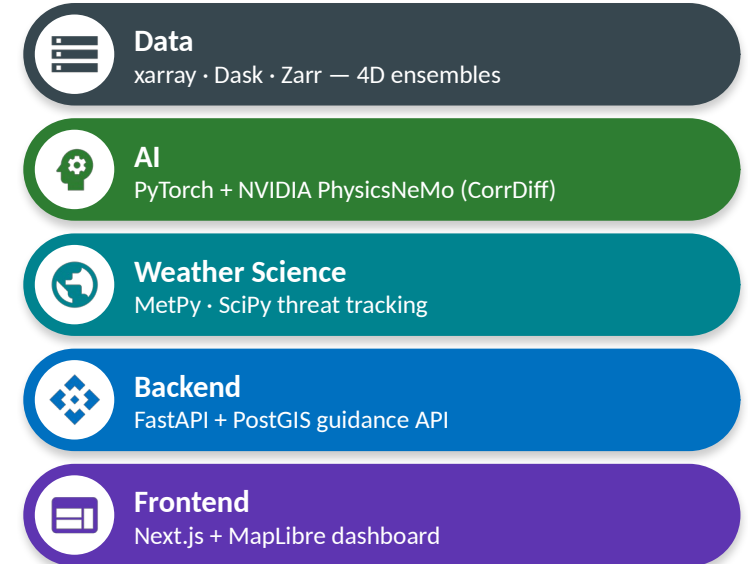
ONE CYCLONE'S JOURNEY Schematic



PROTOTYPE



TECHNOLOGIES USED



Sources: NVIDIA PhysicsNeMo docs (CorrDiff) · ECMWF Forecast User Guide §8.1.9 · NCMRWF/WMO training, 2022 | Output is guidance for IMD forecasters, not public alerts



DATA

IMDAA open on registration · NEPS-G/R via NCMRWF



TOOLS

100% open-source: PyTorch, PhysicsNeMo, xarray



COMPUTE

Pilot-scale training ≈ 10 A100 GPU-hours



SCOPE

1 region · 3 variables · Cyclone Amphan replay



TECHNICAL FEASIBILITY

Analysis of the feasibility of the idea



Proven methods, not new theory

EFI + Shift of Tails is ECMWF operational extreme-weather method. CorrDiff diffusion downscaling is published and open-source.



The data already exists

NEPS-G: 12 km, 23 members, 10 days. NEPS-R: 4 km, 12 members, ~75 hour training target. IMDAA: 12 km, 1979–2020.



Fits the forecast cycle

Processes each 00/12 UTC NEPS-G run automatically after it completes.



Measurable from day one

Standard ensemble scores: CRPS, Fractions Skill Score, power spectra, reliability.



RISKS & MITIGATION

Potential challenges · strategies for overcoming them

Risk	Level	Mitigation strategy
Ensemble archives are not public	High	Formal data request via SIH SPOC in week 1; fallback to IMDAA, clearly disclosed
Trained on days 0–3, applied to days 4–10	High	Lead-time conditioning; coarsened output must match input at every lead
Diffusion may not beat simple statistics	Med	Quantile-mapping baseline runs in parallel; results reported honestly
Cyclone intensity under-corrected	Med	Known CorrDiff limit; we report radius-of-max-wind and peak-wind error
False precision at long lead times	Med	Footprint sized from Fractions Skill Score — never a fixed 5 km pin
NEPS-G model upgrades change inputs	Low	Model version-pinned; input-drift monitor triggers retraining



VIABILITY & DEPLOYMENT

Path from pilot to operations



PHASE 1 · SIH

1 region, 3 variables, Cyclone Amphan replay



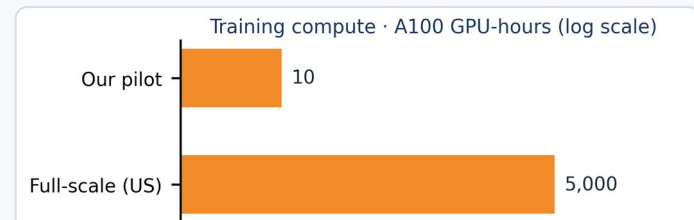
PHASE 2 · Pilot

Multiple regions; open AIFS ensemble as 2nd input



PHASE 3 · Operational

All-India on MoES HPC; feeds IMD impact-based forecasts



Owner: NCMRWF runs it · IMD forecasters use it · open-source, MoES-owned

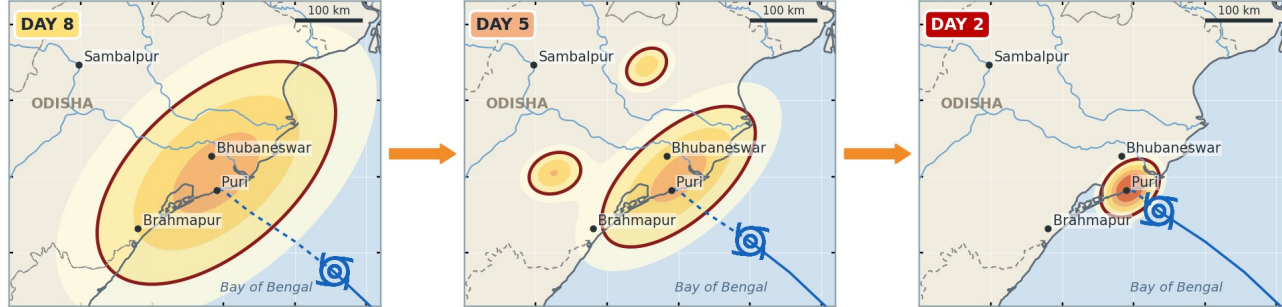
Sources: ECMWF Forecast User Guide §8.1.9 · NVIDIA PhysicsNeMo CorrDiff docs · Mardani et al., Commun. Earth Environ. 2025 · NCMRWF/WMO training, 2022

IMPACT AND BENEFITS



HONEST ALERTS: THE FOOTPRINT SHRINKS AS SKILL RISES

Illustrative alert zones on real geography (Natural Earth) · shading = relative risk, red line = alert footprint



Regional risk zone

Only large scales are predictable

Exposed terrain highlighted

Hills amplify rain in members

Tight local footprint

High-res skill near landfall

Why it matters: a fixed 5 km pin at day 8 would promise precision the atmosphere doesn't allow. Sizing the alert by proven skill (Fractions Skill Score) keeps warnings sharp and credible.



IMPACT ON TARGET AUDIENCE

Potential impact on the target audience · today → with MEGHA-DRISHTI



IMD duty forecaster

Scans ensemble charts by eye → **Ranked threats with evidence, every cycle**



Disaster managers (via IMD)

District-wide warnings → **Terrain-aware zones to pre-position teams**



Farmers (via agromet advisories)

Coarse 12 km guidance at days 4–10 → **Local frost, hail & heat risk for crop decisions**



NCMRWF scientists

No high-res ensemble after day 3 → **AI guidance to day 10 + automatic verification**

BENEFITS OF THE SOLUTION

social · economic · environmental · scientific



SOCIAL

Targeted, credible warnings people can trust and act on



ECONOMIC

GPU inference instead of running extra 4 km ensembles to day 10



ENVIRONMENTAL

Earlier local warning of heat domes, cold waves & cyclones



SCIENTIFIC

High-res ensemble guidance for India, with built-in verification

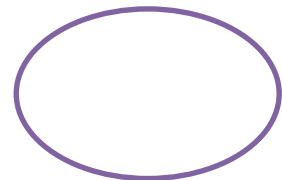
VERIFIED

NEPS-R 4 km ensemble stops at ~75 h · NEPS-G 12 km runs to 10 days · scales below 100 km are predictable for less than a day

TO VALIDATE

Fewer over-broad warnings · forecaster time saved · skill of 5 km guidance at days 4–10 (measured in our pilot)

Sources: NCMRWF/WMO training, 2022 · Weng et al., JGR Atmospheres 2026 · Taraphdar et al., JGR 2014 · MoES Rajya Sabha reply, Feb 2026



RESEARCH AND REFERENCES

Details / Links of the reference and research work · every entry below is clickable

INDIAN SYSTEMS

1

NCMRWF ensemble systems for cyclones

WMO training, New Delhi · 2022

[LINK ↗](#)

2

IMDAA regional reanalysis

Rani et al. · J. Climate · 2021

[LINK ↗](#)

3

MoES reply on AI & impact-based forecasting

Rajya Sabha · Feb 2026

[LINK ↗](#)

METHODS

1

Extreme Forecast Index & Shift of Tails

ECMWF Forecast User Guide §8.1.9

[LINK ↗](#)

2

Physical constraints in ML forecasts (AIFS)

ECMWF · arXiv 2509.18994 · 2025

[LINK ↗](#)

3

Verification of 12 km NEPS

Atmospheric Research · 2019

[LINK ↗](#)

AI MODELS

1

CorrDiff: diffusion km-scale downscaling

Mardani et al. · Commun. Earth Environ. · 2025

[LINK ↗](#)

2

GenCast: diffusion ensemble forecasting

Price et al. · Nature · 2025

[LINK ↗](#)

3

PhysicsNeMo CorrDiff code

NVIDIA · GitHub

[LINK ↗](#)

DATA

1

IndiaWeatherBench (IMDAA, ML-ready)

arXiv 2509.00653 · 2025

[LINK ↗](#)

2

IPED 0.1° rainfall ensemble

Scientific Data · 2025

[LINK ↗](#)

3

NCMRWF reanalysis data portal

NCMRWF · IMDAA & NGFS

[LINK ↗](#)

PREDICTABILITY

1

Scale-dependent predictability limits

Weng et al. · JGR Atmospheres · 2026

[LINK ↗](#)

2

Indian Ocean cyclone predictability

Taraphdar et al. · JGR · 2014

[LINK ↗](#)

FULL DOSSIER



Scan for:
Detailed technical report
Code & prototype
All source links

KEY FINDINGS THAT SHAPED OUR DESIGN *what the research says → what we built because of it*

